

CCT360H5 F

Web Development and Design II

Fall 2026 Syllabus

Course Meetings

CCT360H5 F

Section	Day & Time	Delivery Mode & Location
PRA0104	Friday, 6:00 PM - 9:00 PM	In Person: TBD

Refer to ACORN for the most up-to-date information about the location of the course meetings.

Course Contacts

Course Website: <https://q.utoronto.ca/courses/440563>

Instructor: Gayan Gamage

Email: gayan.gamage@utoronto.ca

Office Hours and Location: By appointment

Additional Notes: Please allow for 24-48 hours for a response. Please include "PRA0104" in the subject line.

Teaching Assistant: Information available in Quercus

Course Overview

This course builds upon the foundations established in CCT260H5 and introduces contemporary standards-based frameworks, tools, and workflows used in the development of responsive front-end systems. Key concepts covered in this course include responsive web design, modern layout techniques, web accessibility, dynamic client-side interactions, frameworks, code version control, web APIs, and introductory server-side implementation concepts. Through a combination of lectures, demonstrations, studio exercises, labs, critiques, and project work, students will develop portfolio-quality web projects using current industry standards and practices.

Course Learning Outcomes

Goals and Learning Objectives

The proliferation of standards, frameworks, and development platforms has created a wide range of opportunities for individuals and organizations to create and publish rich digital experiences. Students engaging in labs, assignments, readings, discussions, and critiques will learn to:

- Develop responsive, accessible, and interactive web applications using HTML5, CSS3, Bootstrap, and JavaScript.
- Plan, prototype, design, and develop web interfaces and dynamic interactions using standards-based frameworks and workflows.
- Apply responsive design principles, modern layout systems, and accessibility standards in the development of digital properties.
- Manipulate and update web content dynamically through JavaScript and DOM scripting techniques.
- Retrieve, process, and display data using web APIs and JSON.
- Understand key concepts related to client-server communication, online publishing, domains, hosting, and introductory server-side workflows.

Prerequisites: CCT260H5

Corequisites: None

Exclusions: None

Recommended Preparation: None

Credit Value: 0.5

Course Materials

Students are required to review all readings, linked resources, tutorials, and materials posted on Quercus throughout the semester.

There is **no required textbook** for this course. Required and supplementary learning materials will be provided through Quercus. Students are encouraged to consult current web development documentation and reference resources, including:

- Mozilla Developer Network (MDN)
- W3Schools
- Bootstrap Documentation
- Web Accessibility Initiative (WAI) Resources

Marking Scheme

Assessment	Percent	Details	Due Date
Labs	30%	Eight lab activities covering responsive design, accessibility, Bootstrap, JavaScript, DOM manipulation, and forms & validation.	Ongoing
Assignment: Project 1	25%	Responsive Professional Website: Design and develop an accessible multi-page responsive website using HTML5, CSS3, and Bootstrap.	2026-10-25

Assessment	Percent	Details	Due Date
Mid Term Test	20%	In-class midterm covering responsive design, accessibility, Bootstrap, and front-end development concepts.	2026-10-23
Assignment: Project 2	25%	Interactive Web Application: Design and develop an interactive web project incorporating JavaScript, dynamic content, user input, and either API integration	2026-12-07

You should receive at least one significant mark (15%) before the last day you can drop a course without academic penalty.

Late Assessment Submissions Policy

Late Penalties

Labs:

Late labs will be penalized **20% per day** for up to **2 days**. Labs submitted more than 2 days late will not be accepted.

Projects 1 & 2:

Late projects will be penalized **10% per day**. Projects submitted more than **7 days** after the deadline will not be accepted.

Requirements and Criteria

Labs

- Eight lab activities will be assigned throughout the semester.
- Each assigned activity is to be completed and submitted to Quercus by the due date posted on Quercus.
- Lab activities will focus on responsive design, accessibility, Bootstrap, JavaScript, DOM manipulation, and forms & validation.

Assignments

- **Project 1** - Responsive Professional Website (Individual Assignment):
Building on the foundations established in CCT260 and the concepts introduced in this course, students will design and develop a professional multi-page responsive website demonstrating accessibility, information architecture, user-centred design, and contemporary front-end development practices using HTML5, CSS3, and Bootstrap.
- **Project 2** - Interactive Web Application (Individual Assignment):

Students will design and develop an interactive web application incorporating responsive design, JavaScript, user interaction, and dynamic content. Projects must incorporate responsive design, JavaScript, user interaction, dynamic content, accessibility considerations, and external API integration

Notes:

- Assignments must be submitted according to the delivery requirements specified in Quercus.
- While the use of online resources is encouraged and recommended, all work submitted must be original. Any code, design assets, templates, frameworks, libraries, or other resources used from external sources must be appropriately cited and adapted to your own work.
- Any media assets, written content, and interface designs should be original work produced by the student unless otherwise specified and cited.
- APA reference style, along with appropriate in-interface citations where applicable, is required. Refer to the section entitled *How Not to Plagiarize* later in this syllabus for instructions on proper citation practices.

Teaching Methods

- This course consists of weekly classes combining lectures, discussions, demonstrations, studio exercises, and hands-on lab activities.
- Students are expected to review required readings and resources, actively practice development techniques, attend classes regularly, participate in activities, and contribute to discussions and critiques.
- Repeated unsupported absences may jeopardize standing in the course.
- Brief reviews of previously covered material may be conducted at the beginning of class; however, one-on-one reviews of missed learning due to absences are not available.
- Course activities emphasize responsive design, accessibility, front-end development workflows, JavaScript programming, and contemporary web technologies through practical project-based learning.

Course Schedule

Week	Description
Week 1 September 7, 2026	• Introduction and Overview: Course Introduction • Modern Web Development Review • Visual Studio Code Setup • GitHub and GitHub Pages Overview • Semantic HTML5 and Accessibility Foundations • Development Tools and Workflow
Week 2 September 14,	• Responsive Design Fundamentals • Mobile-First Design

2026	• Flexbox Layout Systems • Lab 1 Assigned and due September 24, 2026 at 11:59pm
Week 3 September 21, 2026	• CSS Grid Layout Systems • Layout Architecture and Design Systems • CSS Variables and Modern Styling Workflows • Lab 2 Assigned and due October 01, 2026 at 11:59pm
Week 4 September 28, 2026	• Web Accessibility Fundamentals • Accessibility Testing and Evaluation • Introduction to Figma • Site Maps and Wireframes • Lab 3 Assigned and due October 8, 2026 at 11:59pm • Project 1 Assigned
Week 5 October 5, 2026	• Bootstrap Foundations • Containers, Grid Systems and Utility Classes • Responsive Components • Lab 4 Assigned and due October 15, 2026 at 11:59pm
Week 6 October 12, 2026	• Bootstrap Components • Navigation, Cards, Forms and Modals • Framework Workflows and Customization • Lab 5 Assigned and due October 22, 2026 at 11:59pm • Project 1: Site Map / Wireframe / Interface Mockup Due October 15, 2026 at 11:59 PM
Week 7 October 19, 2026	• Midterm test (in Class) • Project 1 Workshop, Q&A and Review • Project 1: Responsive Professional Website Due October 25, 2026 at 11:59 PM
October 26, 2026	Reading Week (no Classes)
Week 8 November 2, 2026	• JavaScript Fundamentals • Variables, Functions, Arrays and Objects • Events and User Interaction
Week 9 November 9, 2026	• DOM Manipulation • Event Listeners • Dynamic Content and Interactive Interfaces • Content Filtering and Search Interfaces

	• Lab 6 Assigned and due November 19, 2026 at 11:59pm • Project 2 Assigned
Week 10 November 16, 2026	• Forms and Validation • Error Handling and User Feedback • Accessibility in Forms • Lab 7 Assigned and due November 26, 2026 at 11:59pm
Week 11 November 23, 2026	• APIs and Dynamic Content • JSON Fundamentals • Fetch API • Working with External Data Sources • Lab 8 Assigned and due December 3, 2026 at 11:59pm • Project 2 Progress Review / Development Check-In
Week 12 November 30, 2026	• Introduction to Node.js • Express Framework • EJS Templates • Client-Server Architecture • Project 2: Interactive Web Application Due December 7, 2026 at 11:59 PM

Policies & Statements

Academic Integrity

The Code of Behaviour on Academic Matters states that:

The University and its members have a responsibility to ensure that a climate that might encourage, or conditions that might enable, cheating, misrepresentation, or unfairness is not tolerated. To this end, all must acknowledge that seeking credit or other advantages by fraud or misrepresentation, or seeking to disadvantage others by disruptive behaviour, is unacceptable, as is any dishonesty or unfairness in dealing with the work or record of a student.

It is your responsibility as a student at the University of Toronto to familiarize yourself with, and adhere to, both the Code of Student Conduct and the Code of Behaviour on Academic Matters. This means, first and foremost, that you should read them carefully.

[Code of Student Conduct](#) and the [Code of Behaviour on Academic Matters](#) are available from the U of T website.

Religious Accommodations

Information about the University's Policy on Scheduling of Classes and Examinations and Other Accommodations for Religious Observances is at

<https://www.vicereprovoststudents.utoronto.ca/student-resources/rights-responsibilities/accommodation-religious/>

Declaration of Temporary Absence

Students who miss an academic obligation during the term (i.e., in-class assessment, quiz, paper or lab report) may use the ACORN Absence Declaration Tool to record an absence in one or more courses. Students may utilize this option once per term for a single absence period of up to seven consecutive days. The declaration period must include the day of declaration and may include past and/or future dates, for a total of up to 7 calendar days.

Use of the ACORN Absence Declaration does not require supporting documentation and should be used in addition to the missed term work policy outlined in the course syllabus. It remains the student's responsibility to initiate the process for missed academic obligations by following the instructions in the course syllabus.

Re-grading Term Work

A student who believes that their written term work has been unfairly marked may ask the person who marked the work for re-evaluation. Students have up to one month from the date of return of an item of term work to inquire about the mark. If the student is not satisfied with this re-evaluation, they may appeal to the instructor in charge of the course if the work was not marked by the instructor (e.g., was marked by a TA). Such re-marking may involve the entire piece of work and may raise or lower the mark. For more information on policies regarding re-marking of term work, please refer to [Re-marking Pieces of Term Work in the Academic Calendar](#).

Robert Gillespie Academic Skills Centre (RGASC)

The Robert Gillespie Academic Skills (RGASC) supports undergraduate and graduate students in their academic skill development (e.g., writing, note taking, time management, study planning, mathematics and numeracy, research). We offer a range of supports, including individual appointments, workshops, and programs (many CCR-accredited). Our programs are designed to help students identify and develop the academic skills they need for success in their studies. The RGASC is located in MN3251 (3rd floor, Maanjiwe nendamowinan building). Explore the RGASC's online resources, book an appointment, and register for one of our programs at <https://www.utm.utoronto.ca/rgasc/>.

University of Toronto Mississauga Library Statement

University of Toronto Mississauga Library. The UTM Library (UTML) is located in the Hazel McCallion Academic Learning Centre (HM). [Open](#) seven days a week, UTML welcomes you to the Information Commons where our public services staff and [liaison librarians](#) offer individual walk-in research assistance, in-person or virtual appointments, workshops, and programs such as the CCR-accredited Library 101 session series focused on helping you develop the

information literacy skills you need to excel in your coursework. Visit the UTM [website](#) to find online books, journals, databases, [research guides](#), [special collections](#) and more information on [on technology-enriched library spaces](#) such as the [Digital Exploration Lab](#) and the [Outer Circle Recording Studio](#).

E-Culture Policy

Only student U of T email accounts should be used for course communication and all emails from students must include the course code and practical section in the subject line and should be signed with the full student name and student number.

It is your responsibility to read your email regularly and check course information updates and announcements through Quercus. Students who choose to opt out of receiving messages through Quercus are still responsible for actions required, or changes communicated through those announcements and messages.

Equity and Academic Rights

The University of Toronto is committed to equity, meaningful inclusion, and respect for diversity. All members of the learning environment in this course should strive to create an atmosphere of mutual respect towards one another. As a course instructor, I will neither condone nor tolerate language or behaviour that undermines the dignity or self-esteem of any individual in this course and wish to be alerted to any attempt to create an intimidating or hostile environment. It is our collective responsibility to create a learning space that is inclusive and welcomes discussion. Discrimination and harassment will not be tolerated. If you have any questions, comments, or concerns I encourage you to bring them to me for us to discuss. You may also contact the UTM Equity, Diversity, and Inclusion Office at edio.utm@utoronto.ca for assistance.

Expectations for Conduct in the Academic Setting

Students agree that by taking this course, they agree to adhere to the ICCIT Expectations for Conduct in the Academic Setting set out at <https://www.utm.utoronto.ca/iccit/student-resources/policies-procedures/iccit-code-conduct>.

Code of Student Conduct

All students are expected to adhere to the Code of Student Conduct ([Code of Student Conduct \[December 13, 2019\] | The Office of the Governing Council, Secretariat](#)).

Accommodation for Missed Tests and Late Assignments

Students must use the new [UTM Special Consideration Request \[Pilot\] application](#) for all ICCIT courses.

Students in CCT109H5, CCT110H5, CCT111H5, CCT112H5 and CCT208H5 should always follow the Special Consideration Request (SCR) process outlined below. Students in other classes should follow the process below only if they are seeking accommodation for tests or assignments worth 20% or more of the final grade. Students in other classes seeking accommodation for tests or assignments worth less than 20% should contact their instructors directly.

Reasons for special consideration could include:

- Accident
- Illness
- Emergency procedure
- Bereavement
- University-sponsored athletics/competitions
- Compulsory legal duties e.g. (jury duty)
- Religious accommodations
- Disability accommodations

Reasons for special consideration do not include*:

- Pre-planned vacations or social commitments
- Transportation delays
- Technology malfunctions
- Time management, course loads Course conflicts, team work conflicts
- Misreading a deadline/timetable
- Late course enrolment
- Scheduled elective medical appointments

* For these situations, refer to your course syllabus and speak directly with your instructor.

You have three days or 72 Hours (including weekends) from the assignment deadline or date of the missed test/quiz to complete the SCR process in full.

The first time in the semester that you are seeking accommodation, please complete the following steps:

1. Login to [ACORN](#), and click on Profile & Settings from the left-hand menu.
2. Click on Absence Declaration
3. Record each day that you are absent – as soon as it begins up until the day you return to campus for classes or other activities.
4. Login to the UTM Special Consideration Request [Pilot] application system and complete the required steps. Documentation is not required. For all subsequent times that you require an accommodation, you should only login to the UTM Special Consideration Request [Pilot] application system and complete the required steps. Documentation is not required, however, it may be requested after the SCR has been reviewed. On your SCR form, please attach a screenshot from Quercus showing the assessment title and deadline. This will help us process your SCR accurately and avoid any misunderstandings. Important note about missed makeup tests: As stated in the [Academic Calendar](#), "If the student is granted permission to take a makeup test and misses it, then they are assigned a mark of zero for the test unless the

instructor is satisfied that missing the makeup test was unavoidable. Students are not automatically entitled to a second makeup test."

If you are registered with Accessibility Services, and the reason for missing a test or a deadline pertains to a disability, you are still required to submit an SCR at the link above. In such cases, the department will accept documentation supplied by Accessibility Services. If you require further information, please speak with your Accessibility Services advisor.

Further details regarding SCR policy are available here:

<https://www.utm.utoronto.ca/iccit/students/student-resources/policies-procedures/special-consideration-requests-scr-late-assignments>

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